

Math 421

Friday, October 31

Announcements

- Homework 7 Due **today**
- Exam 2 – 11/12 5:45-7:15 pm
 - Email me & fill out the form if you have a conflict and need to take the makeup.
 - Sign up for McBurney exam if applicable.

$$f: \mathbb{R} \rightarrow \mathbb{R}$$

largest subset of $\mathbb{R}$
on which f is defined

Activity – Find the natural domains

$$1. \quad f(x) = \frac{1}{x+5}$$

$$\text{dom}(f) = \{x \in \mathbb{R} : x \neq -5\}$$

$$2. \quad g(x) = \sqrt{2 - x^2}$$

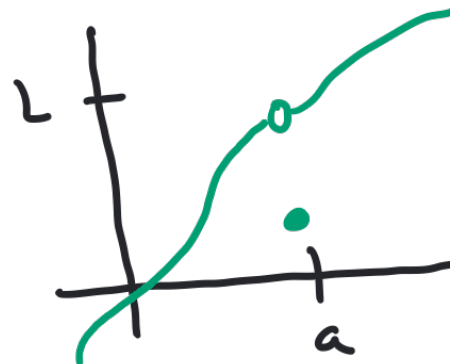
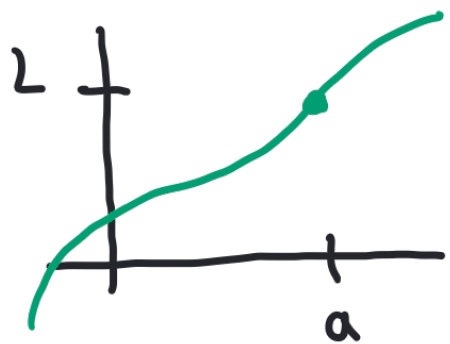
$$\text{dom}(g) = \{x \in \mathbb{R} : -\sqrt{2} \leq x \leq \sqrt{2}\}$$

$$3. \quad \csc x = \frac{1}{\sin x}$$

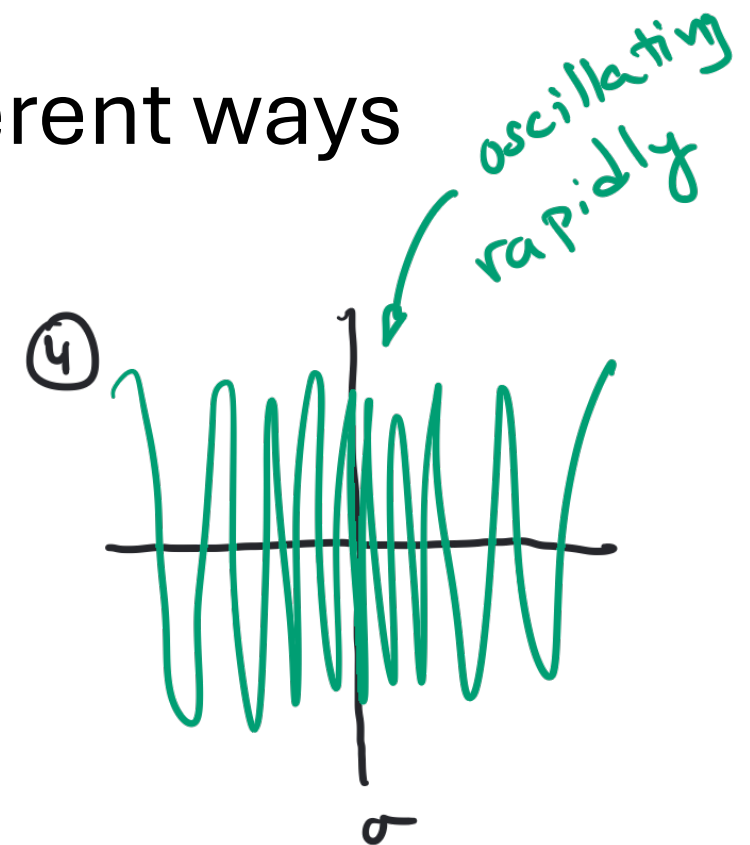
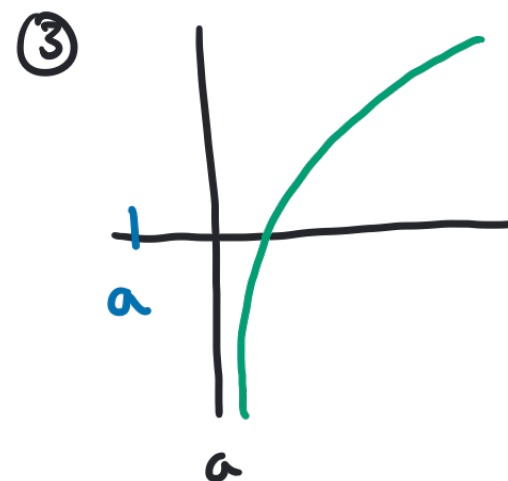
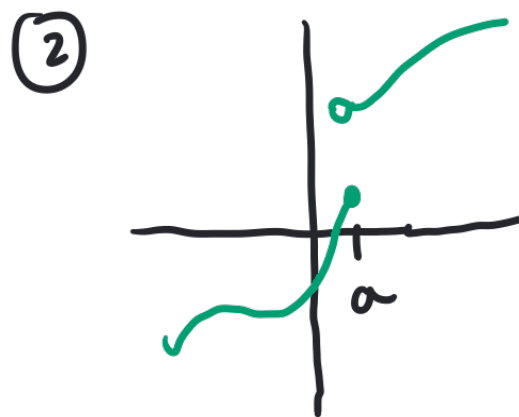
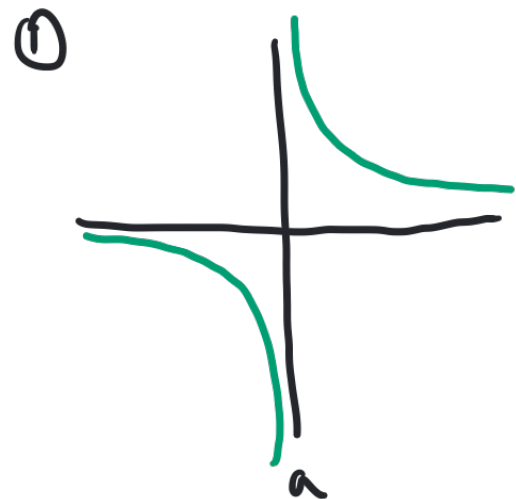
$$\text{dom}(\csc) = \mathbb{R} - \{n\pi : n \in \mathbb{Z}\}.$$

Activity

Recall: $\lim_{x \rightarrow a} f(x) = L$ means that the values of $f(x)$ are close to L when the values of x are close to, but not equal to, a .



Sketch, or provide examples of, at least two different ways $\lim_{x \rightarrow a} f(x)$ can not exist.



Functional Limits – via $\epsilon - \delta$

Def. Let $f: S \rightarrow \mathbb{R}$ and let a be the limit of some sequence in S .

Then $\lim_{x \rightarrow a} f(x) = L$ if for all $\epsilon > 0$ there exists some $\delta > 0$ such that:

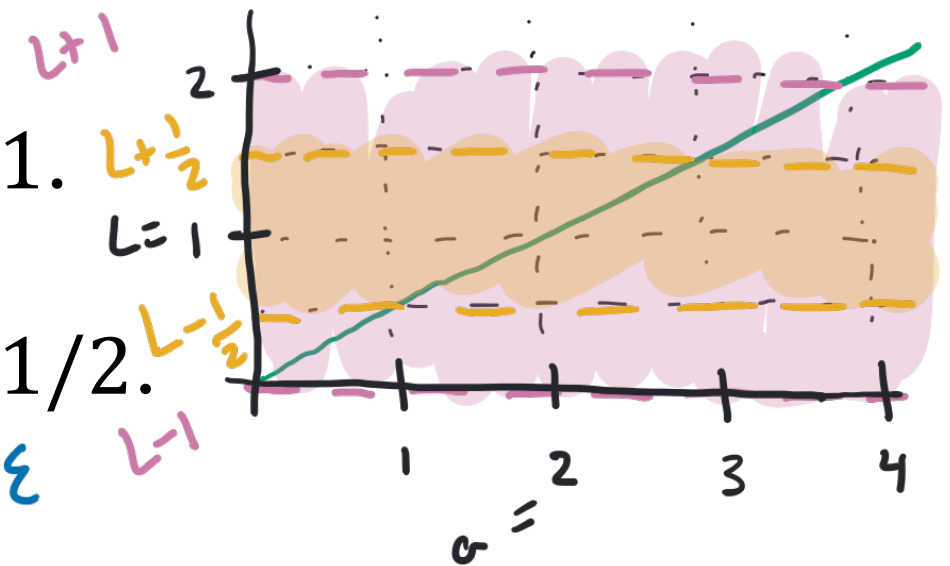
if $0 < |x - a| < \delta$ and $x \in S$, then $|f(x) - L| < \epsilon$.

Activity – Sketch and find δ

1. Let $f(x) = \frac{1}{2}x$, $a = 2$, $L = 1$.

○ If $0 < |x - 2| < \underline{2}$, then $|f(x) - 1| < 1$.

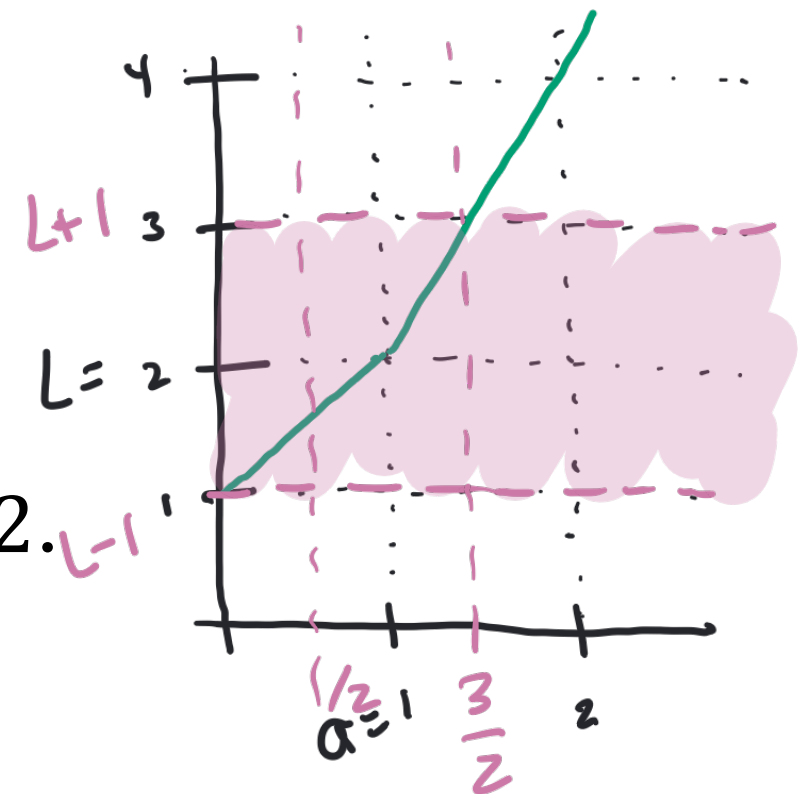
○ If $0 < |x - 2| < \underline{1}$, then $|f(x) - 1| < 1/2$.



2. Let $f(x) = \begin{cases} x + 1, & x \leq 1 \\ 2x, & x > 1 \end{cases}$, $a = 1$, $L = 2$.

○ If $0 < |x - 1| < \underline{1/2}$, then $|f(x) - \mathbf{2}| < 1$.

○ If $0 < |x - 1| < \underline{1/4}$, then $|f(x) - \mathbf{2}| < 1/2$.



$\epsilon - \delta$ Proof Recipe

To show that $\lim_{x \rightarrow a} f(x) = L$.

1. “Fix $\epsilon > 0$.”
2. Determine δ , requires scratch work
3. “Let $\delta = \underline{\hspace{2cm}}$ ” *often involves ϵ .*
4. “Assume $0 < |x - a| < \delta$.”
5. Show that $|f(x) - L| < \epsilon$.
6. “Since $\epsilon > 0$ was arbitrary, $\lim_{x \rightarrow a} f(x) = L$.”

Example

Show $\lim_{x \rightarrow 2} 6x = 12$.

looking to express
 $|f(x) - L|$ in a way
that involves
 $|x - a|$

Discussion. $|6x - 12| < \varepsilon$ whenever $|x - 2| < \delta$

$$|6x - 12| = 6|x - 2| < 6\delta < \varepsilon \Rightarrow \delta = \frac{\varepsilon}{6}.$$

Proof. Let $\varepsilon > 0$. Set $\delta = \frac{\varepsilon}{6}$. Assume $0 < |x - 2| < \delta$. Then

$$|6x - 12| = |6(x - 2)| = 6|x - 2| < 6 \cdot \delta = 6 \cdot \left(\frac{\varepsilon}{6}\right) = \varepsilon.$$

Since $\varepsilon > 0$ was arbitrary, $\lim_{x \rightarrow 2} 6x = 12$.

Activity – Fill in the Blanks

Thm. $\lim_{x \rightarrow 2} 5x = 10.$ $\lim_{x \rightarrow 2} kx = k \cdot 2$ $\lim_{x \rightarrow a} kx = k \cdot a$

Proof. Fix $\epsilon > 0$. Let $\delta = \underline{\epsilon/5}$. Assume $0 < |x - 2| < \delta$. Then

$$|5x - 10| = \underline{|5(x - 2)| = 5|x - 2| < 5 \cdot \delta = 5 \cdot \left(\frac{\epsilon}{5}\right) = \epsilon}$$

_____ *Q.E.D.*

Since $\epsilon > 0$ was arbitrary, $\lim_{x \rightarrow 2} 5x = 10$.